Escape speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain escape in an ideal gravitational-energy model.

SUCCESS CRITERIA - I can...

1. State the zero-at-infinity energy convention.
2. Distinguish escape speed from circular-orbit speed.
3. Name neglected effects in the ideal calculation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “escape speed” and “orbit” to explain your model or result. Sentence starters are optional.

escape speed: minimum ideal speed to leave without further thrust

orbit: a path around another body under gravity

radius: distance from the center to the edge of a circle or sphere

01 Bring back an earlier idea

A 3 kg cart at 10 m/s hits an identical cart at rest. Final velocities are 4 and 6 m/s.
Check momentum and energy; classify the collision.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $v_{escape} = \sqrt{(2\ GM/r)}$; $v_{circular} = \sqrt{(GM/r)}$.
Ideal spherical body; ignore atmosphere/rotation. Minimum escape reaches infinity at zero speed with no further thrust.

Worked reasoning: For a toy planet with GM=50 m^3/s^2 and radius 4 m, escape speed is $\sqrt{(2\ GM/R)}$=5 m/s. At that limiting speed the ideal final speed at infinity is zero.

Your turn: Draw initial kinetic and gravitational-potential-energy bars whose total is zero.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

Does escaping mean that gravity turns off at the launch point?
Explain what the speed allows in an energy model.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

04 Solve and justify

For GM=400 m^3/s^2 and r=8 m, calculate circular speed and escape speed.
Compare their ratio.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Escape speed

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Why is an ideal escape-speed calculation not a complete rocket-launch plan?

Word help: ideal = using stated simplifying conditions

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

06 Choose a method

A 6 kg object descends 8 m near Earth.
Use g=10.
Find ΔU and the work done by gravity.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

At fixed GM, launch distance increases by factor 4.
Compare escape speeds.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A planet's mass doubles while radius stays fixed.
Compare escape speed. Then double both mass and radius and compare again.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3